Response to Reviewers — Paper #518

OrdinalFed: Federated Learning with Ordinal-Aware Aggregation for Diabetic Retinopathy Grading

We thank both reviewers for their careful reading and helpful comments. Each point is answered below. Section, table and figure numbers refer to the revised manuscript.

Reviewer #2: *"... stronger experimental validation through ablation studies, additional recent baselines, independent validation data, repeated experiments, and detailed class-wise performance analysis..."*

Request	What we changed	Where
Ablation studies	We removed one component at a time, under $\alpha = 0.1$ and the real-site partition: OrdinalLoss with image-count weights, cross-entropy with QWK weights, cross-entropy with equal weights, and the full method. We also tested three settings of OrdinalFed (λ, β_0, τ), fixed before the runs and compared on validation data only. The ablation shows that the aggregation, not OrdinalLoss, prevents FedAvg's collapse, and that equal client weights work at least as well as QWK weights.	Sec. IV-D, V-C, Table III
Additional baselines	We added AccWt, a quality-weighted baseline based on server-side validation scoring [9]. It is identical to our aggregation except that clients are scored by accuracy instead of QWK, so it shows whether the ordinal score itself matters. It performs almost the same as QWK weighting. We also added OrdinalFed + Prox, which combines our aggregation with FedProx's proximal term.	Sec. IV-C, Tables I–II
Independent validation data	All main results now come from the DDR test split (3,709 images, excluding the 50 server-set images). These images are never used for training, aggregation or choosing the reported round, which is selected on the validation split. All models score lower on the test split; this is now discussed in the Limitations. In addition, every trained model was scored, without retraining, on the external APTOS 2019 dataset (3,662 images), where all models reach QWK 0.60–0.66, close to the centralised model (0.641).	Sec. IV-A, IV-D, V-E, Tables I, V
Repeated experiments	Every configuration was run with three random seeds; results are reported as mean $\pm$ standard deviation. With three seeds, the late FedProx drop seen in the single run of the submitted version did not recur.	Tables I–V
Class-wise analysis	We now report per-grade recall and F1 on the test split for FedAvg, FedProx and OrdinalFed. No method detects Grade 1 reliably (recall ≤ 0.05); OrdinalFed has the highest Grade 3 recall in both partitions.	Sec. V-D, Table IV

How the conclusions changed: the repeated test-split experiments requested by Reviewer #2 changed our conclusions, and the abstract, introduction, results and conclusion now state them as measured. OrdinalFed still prevents most of FedAvg's collapse under extreme skew (worst-round test QWK 0.514 versus 0.378), but the gain comes from giving small hospitals more weight rather than from the QWK scores or OrdinalLoss, and FedProx is the most stable method. We believe this is a more useful result for the community than the single-run comparison in the submitted version.

Other corrections: Eqs. (4)–(6) now describe exactly the aggregation used in the experiments. The analysis of why the aggregation is stable (Sec. V-C) was rewritten from the per-round server scores and aggregation weights logged during training. The server-side overhead is now the measured value (1.2 s per round), and the dataset statistics in Sec. IV-A were corrected.

Reviewer #3

Comment	What we changed	Where
1. Multiple consecutive reference format, e.g. [1-2]	Consecutive citations are now written as a range (e.g. [5]–[6]), and the references are numbered in the order they are first cited.	Sec. I, Refs.
2. Tiny font inside Figs. 1–2	Both figures were redrawn at their printed size, so all text is at least 8 pt.	Figs. 1–2
3. Description missing for Figs. 1–2	Each figure is now introduced in the text, explaining what its panels, bars and lines show.	Secs. IV-B, V-B
4. IEEE citation format; DOI required	We checked every reference against Crossref and arXiv, rewrote the list in IEEE style and added DOIs (a URL and access date for the IDF Atlas, which has none). Wrong venues, author lists, volumes and pages were fixed.	References
5. Passive voice preferred	The text was revised to use the passive voice.	Throughout
6. Description before all figures and tables	Every figure and table is now mentioned and described in the text before it appears; Figs. 1–2 were moved, and floats are no longer placed above the text that introduces them.	Throughout